



SIB GRAPI

39TH CONFERENCE ON
GRAPHICS, PATTERNS
AND IMAGES 2026

GOIÂNIA - GO 📍 2026

REALIZAÇÃO:



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A Benchmark for Vehicle Attribute Classification in Cross-Domain Surveillance Scenarios

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Optimizing Vehicle Detection

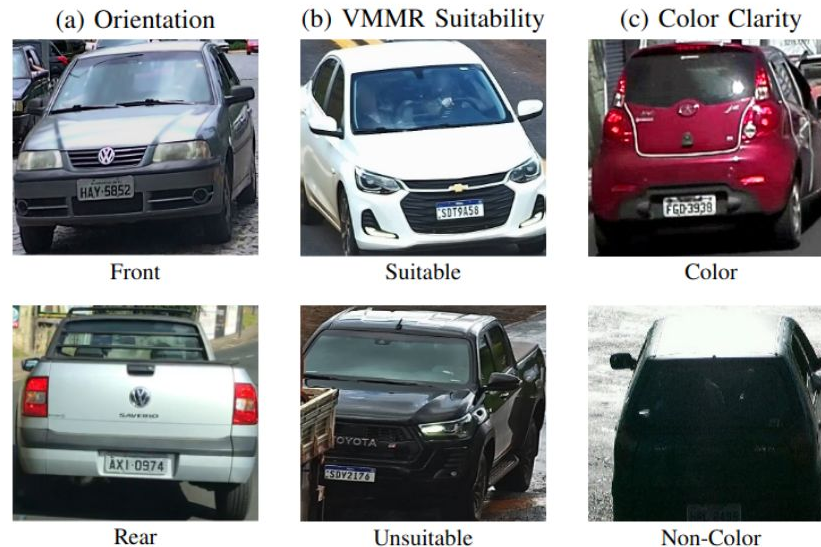
Viewpoint, occlusion, illumination and sensor characteristics vary **substantially** in urban monitoring scenarios and can degrade Intelligent Transportation Systems (ITS) models evaluated under controlled conditions.

To address this gap, we introduce the **Unconstrained Vehicle Identification Benchmark (UVIB)**.



Contributions

- **UVIB** creation and release, with annotations for **three operational vehicle-analysis tasks** across **seven public datasets**;
- Standardized baselines for **four deep architectures** under **four evaluation protocols**;
- **Domain shift analysis**.



Examples of the three target tasks in the UVIB.



The UVIB Benchmark



Vehicle-Rear

LPLCv2

UFPR-VeSV

(a) Surveillance Domain



UFOP

SSIG-SegPlate

UFPR-ALPR

RodoSol-ALPR

(b) General Domain

84.835 vehicle crops from 7 public brazilian datasets. 57.798 from the **Surveillance Domain**, 27.037 from de **General Domain**.



The Surveillance Domain

	Orientation		VMMR Suitability		Color Clarity	
Source Datasets	Front	Rear	Suitable	Unsuitable	Color	Non-Color
Vehicle-Rear (2021)	0	730	641	89	390	340
LPLCv2 (2026)	2,816	29,307	24,413	7,710	19,046	13,077
UFPR-VeSV (2026)	11,103	13,842	23,490	1,455	18,612	6,333
Total Per Class	13,919	43,879	48,544	9,254	38,048	19,750



The General Domain

	Orientation		VMMR Suitability		Color Clarity	
Source Datasets	Front	Rear	Suitable	Unsuitable	Color	Non-Color
UFOP (2011)	298	173	362	109	408	63
SSIG-SegPlate (2016)	2,066	0	1,990	76	1,715	351
UFPR-ALPR (2018)	0	4,500	4,392	108	2,739	1,761
RodoSol-ALPR (2022)	9,989	10,011	19,652	348	8,451	11,549
Total Per Class	12,353	14,684	26,396	641	13,313	13,724



Experimental Setup

Four widely adopted image-classification architectures:

- EfficientNetV2-S;
- ResNet-50;
- Vision Transformer (ViT)/B-16;
- YOLO11s-cls.

Four evaluation protocols:

- Surveillance to General (S2G);
- General to Surveillance (G2S);
- All-Datasets;
- Cross-Dataset Shift (CDS).

Each experiment was repeated with **three random seeds**.
Accuracy, per-class F1-Score (F1) and **Macro F1** are reported
as **mean** and **standard deviation**.



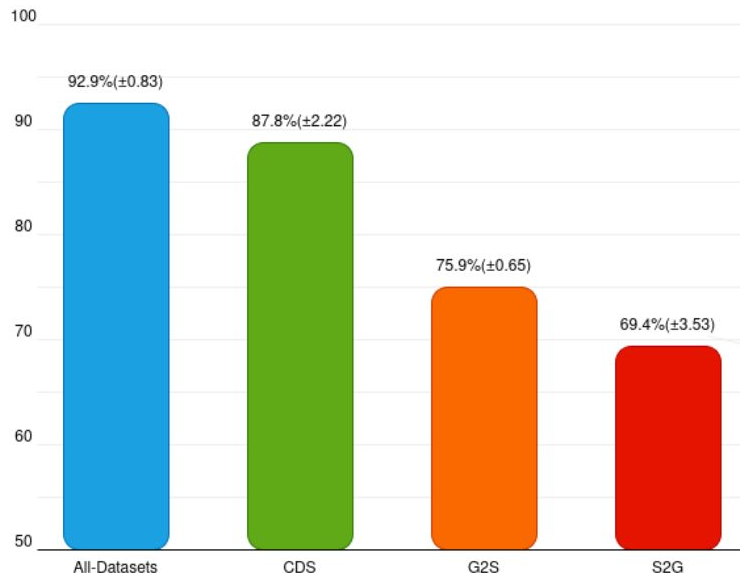
Results

The **protocol** has a stronger effect on performance than the **architecture**.

The **CDS** results in general **outperform** S2G and G2S, while **underperforms** the All-Datasets.

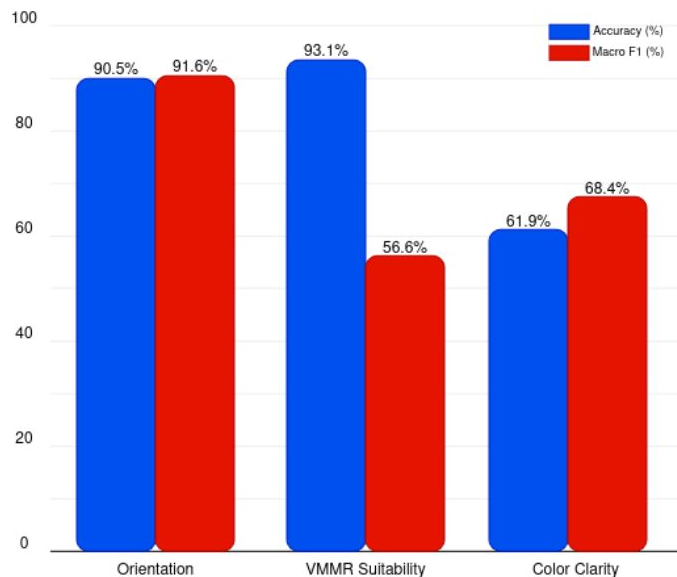
The **All-Datasets** protocol has lower variance compared to the other 3.

Ex: VMMR Suitability - EfficientNetV2-S (Macro F1)



Results

Ex: S2G (YOLO11s-cls) - Accuracy and Macro F1



- **Orientation** is the most **transferable** task;
- **VMMR Suitability** is the most **imbalanced** task;
- **Color Clarity** is **sensitive** to acquisition bias, but more **balanced** than VMMR Suitability;
- **EfficientNetV2-S** is the **strongest** default architecture.



Classification Errors



(a) Orientation



(b) VMMR Suitability



(c) Color Clarity



Future works

1

ANNOTATION FRAMEWORK

Refine the annotation framework by introducing multiple occlusion-severity levels and richer color-quality categories.

2

DOWNSTREAM PIPELINES

Incorporate these tasks into VMMR and VCR pipelines to assess whether filtering or joint prediction improves end-task reliability.

3

IMPROVING RESULTS

Experiments with multi-task learning, synthetic data generation and targeted augmentation strategies to improve rare-condition coverage and reduce cross-domain degradation.





Thank you!

Questions?

